Questions

Q1.

A student was asked to prove by exhaustion that

if n is an integer then $2n^2 + n + 1$ is **not** divisible by 3

The start of the student's proof is shown in the space below.

Consider the case when $n = 3k$

$$2n^2 + n + 1 = 18k^2 + 3k + 1 = 3(6k^2 + k) + 1$$

which is not divisible by 3

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Complete this proof.

(Total for question = 4 marks)

(Q10 WMA12/01, Jan 2023)

Q2.

Given $n \in \mathbb{N}$, prove, by exhaustion, that $n^2 + 2$ is not divisible by 4.

(Total for question = 4 marks)

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(Q04 WMA12/01, Specimen papers)

Q3.

(i) Use algebra to prove that for all real values of x

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$(x - 4)^2 \geq 2x - 9$

(3)

(ii) Show that the following statement is untrue.

$2^n + 1$ is a prime number for all values of n, $n \in \mathbb{N}$

(1)

(Total for question = 4 marks)

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(**Q03 WMA12/01, June 2019**)

Q4.

(i) Use algebra to prove that for all $x \geq 0$

$$3x + 1 \geqslant 2\sqrt{3x}$$

(3)

(ii) Show that the following statement is not true.

"The sum of three consecutive prime numbers is always a multiple of 5"

(1)

(Total for question = 4 marks)

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(**Q05 WMA12/01, Jan 2021**)

Q5.

(i) Prove that for all single digit prime numbers, p,

$p^3 + p$ is a multiple of 10

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(2)

(ii) Show, using algebra, that for $n \in \mathbb{N}$

$$(n + 1)^3 - n^3 \text{ is not a multiple of 3}$$

(3)

(Total for question = 5 marks)

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(**Q03 WMA12/01, June 2021**)

Q6.

(i) Prove by counter example that the statement

"if p is a prime number then $2p + 1$ is also a prime number"

is not true.

(1)

(ii) Use proof by exhaustion to prove that if n is an integer then

$$5n^2 + n + 12$$

is always even.

(4)

(Total for question = 5 marks)

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(**Q10 WMA12/01, Jan 2022**)

Q7.

An arithmetic series has first term a and common difference d.

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(a) Prove that the sum of the first n terms of the series is

$$\tfrac{1}{2}n[2a + (n-1)d]$$

(4)

A company, which is making 200 mobile phones each week, plans to increase its production.

The number of mobile phones produced is to be increased by 20 each week from 200 in week 1 to 220 in week 2, to 240 in week 3 and so on, until it is producing 600 in week N.

(b) Find the value of N

(2)

The company then plans to continue to make 600 mobile phones each week. RocketRevise

(c) Find the total number of mobile phones that will be made in the first 52 weeks starting from and including week 1.

(5)

(Total for question = 11 marks)

(**Q05 WMA12/01, Specimen papers**)

Q8.

Given that a, b and c are integers greater than 0 such that

- $c = b + 2$
- $a + b + c = 10$

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Prove, by exhaustion, that the product of a, b and c is always even.

You may use the table below to illustrate your answer.

(3)

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You may not need to use all rows of this table.

a	*b*	*c*
	1	
	2	

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(Total for question = 3 marks)

(Q01 WMA12/01, Oct 2022)

Q9.

Given that a, b and c are integers greater than 0 such that

- $c = 3a + 1$
- $a + b + c = 15$

prove, by exhaustion, that the product abc is always a multiple of 4

You may use the table below to illustrate your answer.

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(3)

You may not need to use all rows of this table.

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a	b	c	abc

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(Total for question = 3 marks)

(Q01 WMA12/01, Oct 2023)

Q10.

(i) A student writes the following statement:

When a and b are consecutive **prime** numbers, $a^2 + b^2$ is never a multiple of 10

Prove by counter example that this statement is **not** true.

(2)

(ii) Given that x and y are even integers greater than 0 and less than 6, prove by exhaustion, that

$$1 < x^2 - \frac{xy}{4} < 15$$

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(3)

(Total for question = 5 marks)

(Q08 WMA12/01, June 2023)

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Q11.

(i) Show that the following statement is **false**:

$$\text{"}(n + 1)^3 - n^3 \text{ is prime for all } n \in \mathbb{N}\text{"}$$

(2)

(ii) Given that the points $A(1, 0)$, $B(3, -10)$ and $C(7, -6)$ lie on a circle, prove that AB is a diameter of this circle.

(5)

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(Total for question = 7 marks)

(Q03 WMA12/01, June 2022)

Q12.

The curve C has equation

$$y = ax^3 - 3x^2 + 3x + b$$

where a and b are constants.

Given that

- the point (2, 5) lies on C
- the gradient of the curve at (2, 5) is 7

(a) find the value of a and the value of b.

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(4)

(b) Prove that C has no turning points.

(3)

(Total for question = 7 marks)

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(Q10 WMA12/01, Oct 2019)

Q13.

A geometric series has first term a and common ratio r.

(a) Prove that the sum of the first n terms of this series is given by

$$S_n = \frac{a(1-r^n)}{1-r}$$

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(3)

The second term of a geometric series is -320 and the fifth term is $\frac{512}{25}$

(b) Find the value of the common ratio.

(2)

(c) Hence find the sum of the first 13 terms of the series, giving your answer to 2 decimal places.

(3)

(Total for question = 8 marks)

(Q08 WMA12/01, Oct 2020)

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Q14.

(i) An arithmetic series has first term α and common difference d.

Prove that the sum to n terms of this series is

$$\frac{n}{2}\{2a + (n-1)d\}$$

(3)

(ii) A sequence u_1, u_2, u_3,... is given by

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$$u_n = 5n + 3(-1)^n$$

Find the value of

(a) u_5

(1)

(b) $\sum_{n=1}^{59} u_n$

(3)

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(Total for question = 7 marks)

(**Q08 WMA12/01, Jan 2020**)

Q15.

(a) Prove that for all positive values of x and y,

$$\frac{x+y}{2} \geqslant \sqrt{xy}$$

(3)

(b) Prove by counter-example that this inequality does not hold when x and y are both negative.

(1)

(Total for question = 4 marks)

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(**Q09 WMA12/01, Oct 2021**)

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